

Carlos Xavier Hernández

Senior Research Scientist

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I am a Senior Research Scientist at [Meta Reality Labs](#), working on machine learning to enable [neuromotor interfaces](#). Prior to that, I worked with [Vijay Pande](#) at Stanford on statistical modeling of biomolecular dynamics.

EXPERIENCE

Meta Platforms, Inc.

Senior Research Scientist · New York, NY, USA · 2019 – Present

- Shipped gesture recognition to consumers as the technical lead of a team of 8+ research scientists and engineers developing for the [Meta Neural Band](#) (launched Sep 2025), by designing and training deep learning models that decode real-time sEMG and IMU signals into discrete input controls for [Meta Ray-Ban Display](#) working under tight hardware constraints.
- Achieved >90% gesture classification accuracy on held-out users without the need for individual calibration, by architecting a generic LSTM-based neural decoding model trained on large-scale sEMG datasets collected from ~5,000 participants. Co-authored [peer-reviewed publication in Nature](#) demonstrating the first high-bandwidth non-invasive neuromotor interface with cross-user generalization (0.88 gestures/sec in closed-loop tests with first-time users), contributing core ML model development and evaluation methodology.
- Demonstrated viability of [EMG-based controls for users with hand tremor](#) (featured at Meta Connect 2024), by leading cross-functional accessibility data collection and analysis to show that EMG-based models can accurately decode motor intent despite involuntary movement artifacts, achieving 80+% gesture classification accuracy on the tremor population.

CTRL-labs (acquired by Meta, 2019)

Research Scientist · New York, NY, USA · 2018 – 2019

- Built production-grade ML training and inference pipelines for personalization of real-time gesture recognition models, by developing end-to-end data processing, training, and fine-tuning infrastructure for wrist-based sEMG decoding.
- Established foundational R&D for EMG-based neural interfaces prior to acquisition, by conducting early research on time-series signal processing and deep learning approaches for decoding motor signals into user intent.

Stanford University

NSF Graduate Research Fellow · Stanford, CA, USA · 2013 – 2018

- Authored [peer-reviewed publication in Phys. Rev. E](#) describing the Variational Dynamical Encoder (VDE), a time-lagged variational autoencoder that compresses high-dimensional time-series into a single interpretable, low-dimensional latent representation. The VDE retained over twice the mutual information with input features compared to linear methods, and on protein folding simulations resolved a slowest dynamical process 2× longer than the leading linear baseline (tICA).
- Co-developed open-source scientific computing tools widely adopted across the computational biology community, by building Python libraries for molecular dynamics trajectory analysis, Markov state modeling of biomolecular kinetics, and automated hyperparameter optimization.
- Achieved an R^2 of 0.987 and MSE of <0.1 in automated cell counting and segmentation, by training a convolutional neural network pipeline (FPN + VGG-11) with uncertainty estimation on ~10,000 microscopy images, replacing a time-intensive manual process with computer vision.

EDUCATION

Stanford University

Ph.D. in Biophysics · Stanford, CA, USA · 2013 – 2018

Advisor: [Vijay Pande](#)

Columbia University in the City of New York

B.S. in Applied Mathematics · New York, NY, USA · 2009 – 2013

SKILLS

Languages & Frameworks: Python, PyTorch, NumPy, SciPy, Pandas

Domains: Time-series modeling, signal processing (DSP), causal inference

Methods: Deep learning (RNNs, Transformers), large-scale distributed training, fine-tuning, Markov state models, information theory

SELECTED PUBLICATIONS

A Generic Non-Invasive Neuromotor Interface for Human-Computer Interaction

P Kaifosh, TR Reardon, and **CTRL-labs** · [Nature](#) · 2025

 63

Variational Encoding of Complex Dynamics

CX Hernández*, HK Wayment-Steele*, MM Sultan*, BE Husic, and VS Pande · [Phys. Rev. E](#) · 2018

 149

Using Deep Learning for Segmentation and Counting within Microscopy Data

CX Hernández, MM Sultan, and VS Pande · [arXiv](#) · 2018

 36

SELECTED SOFTWARE

MDTraj: A Modern, Open Library for the Analysis of Molecular Dynamics Trajectories

RT McGibbon, KA Beauchamp, MP Harrigan, C Klein, JM Swails, **CX Hernández**, CR Schwantes, LP Wang, TJ Lane, and VS Pande · [mdtraj/mdtraj](#)

Python ·  689 ·  290

VDE: Variational Dynamical Encoder for Complex Dynamics

CX Hernández, HK Wayment-Steele, MM Sultan, BE Husic, and VS Pande · [msmbuilder/vde](#)

Python ·  189 ·  42

MSMBuilder: Statistical Models for Biomolecular Dynamics

MP Harrigan, MM Sultan, **CX Hernández**, BE Husic, P Eastman, CR Schwantes, KA Beauchamp, RT McGibbon, and VS Pande · [msmbuilder/msmbuilder](#)

Python ·  161 ·  94

MolEncoder: Molecular Autoencoder in PyTorch

CX Hernández · [cxhernandez/molencoder](#)

Python ·  92 ·  18

Osprey: Hyperparameter Optimization for Machine Learning

RT McGibbon, **CX Hernández**, MP Harrigan, S Kearnes, MM Sultan, S Jastrzebski, BE Husic, and VS Pande · [msmbuilder/osprey](#)

Python ·  73 ·  26

POSTERS & PRESENTATIONS

Neural Control of Movement Society Meeting

Poster · Panama City, PAN · 2025 “Stable Control through sEMG Input: Hand Gesture Recognition on a Population with Hand Tremor”

Convolutional Neural Networks for Visual Recognition (CS231N)

Invited Presentation · Stanford, CA, USA · 2017 “Using Deep Learning for Segmentation and Counting within Microscopy Data”

Biophysical Society Meeting

Poster · Los Angeles, CA, USA · 2016 “Intrinsic Disorder in the P53 C-Terminal Regulatory Domain Yields Multiple Pathways for Folding-Upon-Binding”

Workshop on Molecular and Chemical Kinetics

Poster · Berlin, DEU · 2015 “Inferring Causality Along Transition State Pathways”

HONORS & AWARDS

- Graduate Research Fellowship · National Science Foundation · 2013
- ADVANCE Summer Research Fellowship · Stanford University · 2013
- EXROP Undergraduate Research Fellowship · Howard Hughes Medical Institute · 2012
- Genentech Summer Undergraduate Research Fellowship · Columbia University · 2011

PRESS

- [A Look at Our Surface EMG Research Focused on Equity and Accessibility](#) · *Meta* · 2024
- [Move Objects With Your Mind? We're Getting There, With The Help Of An Armband](#) · *NPR* · 2019
- [Pandemic Flu Risk Raised by Lax Hog-Farm Surveillance](#) · *Wired* · 2012